

CONDITIONAL EXPECTATION

LAST WEEK: CONDITIONAL PMFs / PDFs

$P_{Y|X}(y|x)$ (Y IS DISCRETE)

$f_{Y|X}(y|x)$ (Y IS CONTINUOUS)

↑ FUNCTION OF x

$$\begin{array}{l} E(Y|X=x) \\ \text{"} \\ E(Y|x) \end{array} = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

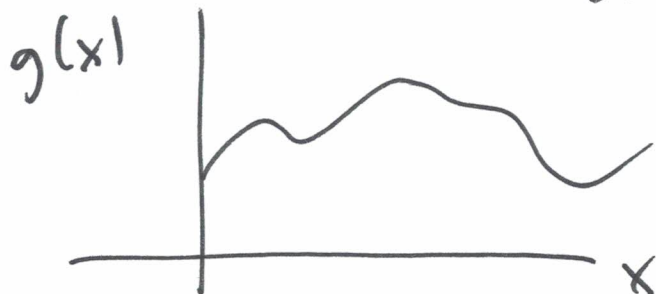
↑ FUNCTION OF x .

WE CAN THINK ABOUT

$$E(Y|X=x) = g(x) \quad \text{AS A}$$

RANDOM VARIABLE ITSELF.

- TELL ME $X=x$
- CREATE CONDITIONAL PDF $f_{Y|X}(y|x=x)$
- COMPUTE CONDITIONAL EXPECTED VALUE $E(Y|X=x)$
- GIVE THAT A NAME $g(x)$



THEN WE COULD ASK, WHAT IS

$$E(g(x))?$$

$$E(g(x)) = \int g(x) f_x(x) dx$$

$$= \int E(Y|x) f_x(x) dx$$

$$= \int \left(\int y \overset{\text{CONDITIONAL}}{f_{Y|x}}(y|x) dy \right) \overset{\text{MARGINAL}}{f_x(x)} dx$$

$$= \iint y \overset{\text{JOINT}}{f_{XY}}(x,y) dy dx$$

$$= \int y \overset{\text{MARGINAL}}{f_Y}(y) dy = E(Y)$$

THAT IS, $E(E(Y|X)) = E(Y)$

THE LAW OF ITERATED EXPECTATIONS

EX $E(\overset{Y}{\cancel{X}} = \text{BLOOD SUGAR}) =$

$$E(\overset{Y}{\cancel{X}} | D) \quad E(\overset{Y}{\cancel{X}} | N)$$



X BINARY

DIABETIC or NOT?

$$= E(Y|D) P(D) + E(Y|N) P(N)$$

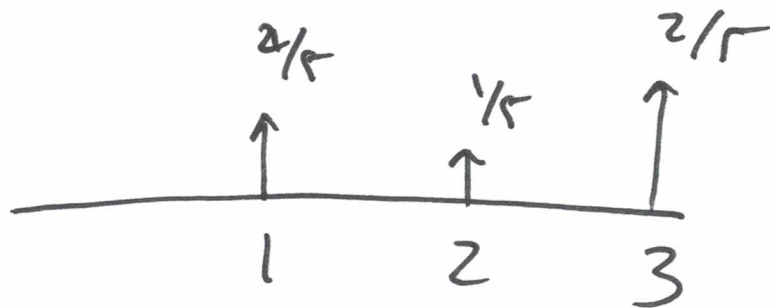
CONSIDER ALYSSA / FIVEFN EXAMPLE:

JOINT PMF:

A \ E						P _A
		0	1	2	3	
A	0	0	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{12}$	$\frac{5}{24}$
	1	$\frac{1}{12}$	0	$\frac{1}{24}$	$\frac{1}{12}$	$\frac{5}{24}$
	2	$\frac{1}{12}$	$\frac{1}{12}$	0	$\frac{1}{12}$	$\frac{6}{24}$
	3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{6}$	0	$\frac{8}{24}$
		$\frac{1}{4}$ $\frac{1}{4}$ $\frac{1.5}{4}$ $\frac{1}{4}$ $\frac{1}{4}$				

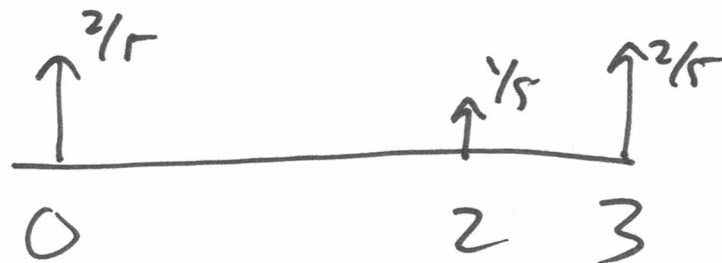
~~$E(E|A=0)$~~

$$E(E_{\substack{| \\ L \\ E \\ N}} | A=0) =$$



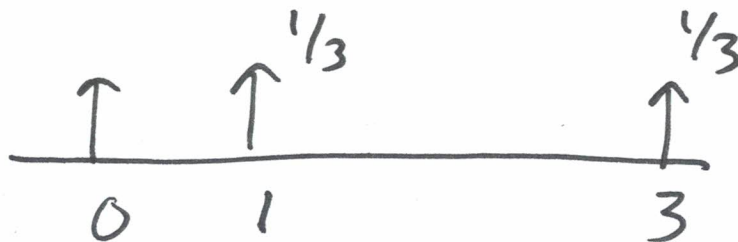
$$\boxed{2}$$

$$E(E_{\substack{| \\ L}} | A=1) =$$



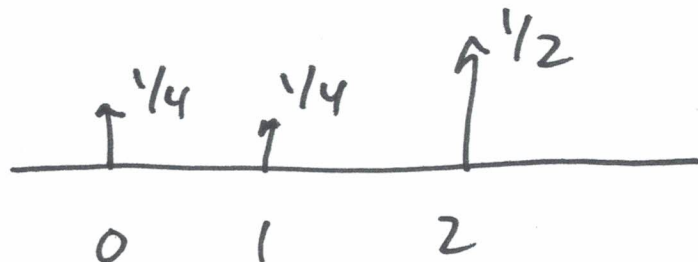
$$\frac{2}{5} + \frac{6}{5} = \boxed{\frac{8}{5}}$$

$$E(E_{\substack{| \\ L}} | A=2)$$



$$\boxed{\frac{4}{3}}$$

$$E(E_{\substack{| \\ L}} | A=3)$$



$$= \boxed{\frac{5}{4}}$$

$$E(E_{\substack{| \\ L}}(E|A)) = 2 \cdot \frac{5}{24} + \frac{8}{5} \cdot \frac{5}{24} + \frac{4}{3} \cdot \frac{6}{24} + \frac{5}{4} \cdot \frac{3}{24}$$

$$\frac{10}{24} + \frac{8}{24} + \frac{8}{24} + \frac{10}{24} = \frac{36}{24} = \boxed{1.5}$$

CONTINUOUS EXAMPLE:

FROM PREVIOUS CLUSTER,

$$f_{XY}(x, y) = \begin{cases} \frac{3}{4}(2x + y^2) & x, y \in [0, 1] \\ 0 & \text{ELSE} \end{cases}$$

LAST TIME, WE FOUND

$$f_{Y|X}(y|\overset{\text{FIXED}}{\underset{\uparrow}{x}}) = \frac{\frac{3}{2}\overset{\circ}{x} + \frac{3}{4}y^2}{\frac{3}{2}\overset{\circ}{x} + \frac{1}{4}} \quad \begin{array}{l} \leftarrow \text{JOINT} \\ \leftarrow \text{MARGINAL IN } x \end{array}$$

$$\begin{aligned} E(Y|X) &= \int_0^1 y \frac{\frac{3}{2}x + \frac{3}{4}y^2}{\frac{3}{2}x + \frac{1}{4}} dy = \frac{1}{\frac{3}{2}x + \frac{1}{4}} \left[\frac{3}{4}y^2x + \frac{3}{16}y^4 \right]_{y=0}^{y=1} \\ &= \frac{\frac{3}{4}x + \frac{3}{16}}{\frac{3}{2}x + \frac{1}{4}} \end{aligned}$$

$$E(E(Y|X)) = \int_0^1 \underbrace{\frac{\frac{3}{4}x + \frac{3}{16}}{\frac{3}{2}x + \frac{1}{4}}}_{\text{FUNCTION OF } X} \cdot \underbrace{\frac{3}{2}x + \frac{1}{4}}_{\text{PDF OF } X} dx$$

$$= \int_0^1 \left(\frac{3}{4}x + \frac{3}{16} \right) dx = \left[\frac{3}{8}x^2 + \frac{3}{16}x \right]_{x=0}^{x=1} = \frac{3}{8} + \frac{3}{16} = \boxed{\frac{9}{16}}$$

THIS TURNS OUT TO BE THE SAME
NUMBER AS

$$\int_0^1 \underbrace{y}_{\text{MARGINAL IN } Y} f_Y(y) dy = \int_0^1 y \cdot \left(\frac{3}{4}y^2 + \frac{3}{4} \right) dy \dots = \frac{9}{16}.$$

IF WE WANTED $E(Y)$:

$$1) \iint y f_{xy}(x,y) dx dy$$

IF WE
HAD JOINT.

$$2) \int y f_Y(y) dy$$

IF WE HAD
MARGINAL

$$3) \int E(Y|x) f_X(x) dx$$

IF WE HAD
CONDITIONAL
EXPECTATION.

SETUP IS REMOTE "POST OFFICE"

INTERARRIVAL TIME FOR VISITORS FROM

TROY: EXPONENTIAL WITH $\lambda_T = \frac{1}{8}$ VISITORS/
WEEK.

INTERARRIVAL TIME FOR VISITORS FROM

SAN FRANCISCO: EXPONENTIAL WITH $\lambda_S = \frac{1}{2}$
VISITORS/
WEEK.

WHAT IS THE EXPECTED # OF (VISITORS FROM)
POSTCARDS TO

TROY DROPPED OFF BETWEEN EACH PAIR

OF SAN FRANCISCO VISITORS?

(PROBABLY: NOT MANY)

T_M : INTERARRIVAL TIME FOR TROY

T_S : INTERARRIVAL TIME FOR S.F.

$$f_{T_M}(t) = \frac{1}{8} e^{-\frac{1}{8}t} \quad t \geq 0$$

$$f_{T_S}(t) = \frac{1}{2} e^{-\frac{1}{2}t} \quad t \geq 0$$

N_T = # OF TROY POSTCARDS B/W
PAIR OF SF VISITORS.

APPROACH: $E(N_T) = E(E(N_T | T_S))$

 **FIXED TIME**

$E(N_T | T_S)$ INVOLVES A POISSON RANDOM
VARIABLE

FIXED
↓

$$P_{N_T | T_S}(n | T_S = t) = \frac{\left(\frac{1}{8}t\right)^n}{n!} e^{-\frac{1}{8}t}$$

POISSON PMF $\alpha = \frac{1}{8}t$.

$$E(N_T | T_S) = \alpha = \frac{1}{8}T_S$$

$$\begin{aligned} E(E(N_T | T_S)) &= \int_0^{\infty} \frac{1}{8}T_S \cdot \frac{1}{2}e^{-\frac{1}{2}T_S} dT_S \\ &= \frac{1}{8}E(T_S) = \frac{1}{8} \cdot 2 = \boxed{\frac{1}{4}}. \end{aligned}$$

SIMILARLY, WE CAN COMPUTE THAT

$$E(N_S \text{ BETWEEN TROY VISITORS}) = \boxed{4}$$

MUCH MORE LIKELY.

$P(\text{A GIVEN POSTCARD FROM TROY IS DROPPED AND
DELIVERED BETWEEN 2 SF VISITORS?})$

$$= P(\geq 2 \text{ TROY VISITORS BETWEEN SF
VISITORS})$$

$$1 - P(N_T = 0) - P(N_T = 1) \dots \frac{1}{25}.$$

CONDITION ON THE UNKNOWN TIME T_S
BETWEEN SF. VISITS.

$$P(N_T = 0 \mid T_S = t)$$

$$P(N_T = 1 \mid T_S = t)$$

$$P(N_T = 0 \mid \text{B/W SF}) = \int P(N_T = 0 \mid T_S = t) f_{T_S}(t) dt$$

TAKE - HOME MESSAGE:

$$E(E(Y|X)) = E(Y)$$

LIKE YOU WOULD (HOPEFULLY) EXPECT:

WEIGHT CONDITIONAL EXPECTED VALUES
THAT DEPEND ON X BY
THE PDF OF X.

GOING FORWARD...

WHAT IS PDF OF $X + Y$?

WHAT ABOUT $X_1 + \dots + X_N$?